Less Architecture, More Recovery: Consensus-Gated Mask-IoU Vote Fusion and Self-Correcting Inference for SAR Ship Instance Segmentation

Bingzhi Hou, *Student Member, IEEE,* et al.

***Abstract*—Deep learning has advanced ship instance segmentation in synthetic aperture radar (SAR) imagery. However, the residual error concentrates in the near-shore regime, where small, low-contrast hulls are embedded in bright dock clutter and specialized architectures lose 25 to 30 mask average precision (AP) points relative to their offshore performance. Existing methods respond with further architectural specialization, while the pretraining regime and the fusion of overlapping candidates go unexamined. This article proposes a consensus-gated mask intersection-over-union (IoU) vote fusion operator, which clusters candidates pooled from augmented views by mask IoU and, gated by cross-view agreement, emits either the seed mask or a consensus-denoised mask per cluster; the weight soup and wide multi-scale flip test-time augmentation (TTA) supplying the views are established practice and act only as its carrier. Two self-correcting stages follow: a boundary-patch refiner confined to a four-pixel band edits mask pixels only, a learned consensus re-ranker edits detection scores only, both fitted on the frozen model's own training-split errors without new annotation. A companion study isolates the pretraining regime. Under our 55-epoch protocol, Common Objects in Context (COCO) initialization lifts an unmodified Mask2Former to 27.7 inshore AP, against the 17.9 reported under the 500-epoch from-scratch-head protocol. Seven architecture- and training-side modifications were flat, negative, or absorbed. The complete cascade reaches 44.5/59.4/70.1 mask AP on the HRSID inshore/official/offshore splits with an inshore AP75 of 48.3 and 62.1/67.5/69.8 on PSeg-SSDD. The training-free stack and the refiner lift six carrier–dataset pairs across five distinct architectures on every split.**

***Index Terms*—Consensus fusion, instance segmentation, self-correction, ship detection, synthetic aperture radar (SAR), test-time augmentation (TTA).**

I. Introduction

WITH its all-weather, day-and-night imaging capability, synthetic aperture radar (SAR) plays a key role in maritime surveillance, fisheries monitoring, and vessel traffic management. Ship instance segmentation extends detection from bounding boxes to pixel-accurate masks and thereby supplies the hull geometry that heading estimation, berthing analysis, and fine-grained recognition require. The release of pixel-annotated benchmarks such as HRSID [1] has made the task quantitatively comparable across methods, and deep segmentation networks now dominate the reported results.

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The authors are with XXX University, XXX, China (e-mail: bingzhi495@gmail.com).

Progress, however, splits between two regimes. Offshore, well-resolved ships on open water are handled adequately by generic instance segmentation models; inshore, small hulls with low or even negative contrast are buried in bright, structured dock clutter under single-look speckle, and the specialized architectures published to date lose 25 to 30 mask average precision (AP) points relative to their own offshore performance, while classic detectors and from-scratch baselines lose more than 30. The open problem therefore lives almost entirely in the near-shore split. The prevailing response has been architectural: an optimized query generator with an orientation-aware embedding module [2], a diffusion-based instance decoder with asymmetric-convolution mask refinement [3], a bounding-box-free elliptical Gaussian-heatmap encoding [4], low-level feature guidance through super-resolution denoising [5], and parameter-efficient transfer of the Segment Anything Model [6]. These works share one protocol, namely, 500-epoch training with an ImageNet backbone and a from-scratch segmentation head, and all attribute their gains to new operators. Two degrees of freedom remain unexamined on that route. Outside instance segmentation, SARDet-100K studies pretraining transfer for box-level SAR detection [7]. Within SAR ship instance segmentation, no published method isolates the pretraining regime as a variable. Nor does any study the fusion step in which overlapping candidates are merged, although that step decides whether the evidence gathered from multiple views of a scene accumulates or cancels.

This article starts from a measurement instead. Under the from-scratch-head protocol, plain Mask2Former [8] scores 17.9 inshore mask AP on HRSID [2]; the identical architecture initialized from Common Objects in Context (COCO) instance segmentation weights reaches 27.7 after 55 epochs under our protocol. A controlled seven-direction sweep on the stronger base finds no architecture- or training-side modification that pays. We therefore leave the model untouched and recover accuracy behind it with a three-rung inference cascade (Fig. 1). The first rung is training-free and is built around a consensus-gated mask intersection-over-union (IoU) vote fusion operator. It clusters the candidates pooled from augmented views by true mask IoU instead of box IoU and, gated by cross-view agreement, emits either the seed mask or a consensus-denoised mask per cluster. The weight soup [9] and the wide multi-scale flip test-time augmentation (TTA) that supply those candidates are established practice and serve only as its carrier.

The remaining two rungs are self-correctors that never retrain or modify the main model. A boundary-patch refiner in the boundary-patch refinement (BPR) family [10] edits mask pixels only, inside a ± 4-pixel band around predicted boundaries, and a learned consensus re-ranker edits detection scores only. Both are fitted on the frozen base model's own training-split errors, and no new annotation is drawn. On HRSID the cascade carries the 27.7-AP base to 44.5 inshore mask AP and 48.3 inshore AP_{75}, above the strongest reported 500-epoch specialized architecture (O2Former, 43.1 and 45.9) at 55 epochs of training and roughly $8\times$ inference cost, although that architecture keeps a 1.6-point offshore lead. On PSeg-SSDD the same recipe reaches 62.1 inshore mask AP, above all six comparison methods we trained under one protocol. Applied without per-carrier tuning to six further carrier–dataset pairs across five distinct architectures, the training-free stack and the refiner improve every pair on every split.

The main contributions of this article are summarized as follows.

- A consensus-gated mask-IoU vote fusion operator that clusters cross-view candidates by mask IoU and, within a cross-view agreement band, chooses between the seed mask and a consensus-denoised mask per cluster. The weight soup and the wide TTA supplying the views serve only as its carrier. On identical candidates the operator adds +1.7/+1.2/+0.6 inshore/official/offshore AP over box non-maximum suppression (NMS), and its value is that it makes wide TTA safe. Widening the view set from six to eight under box-NMS lowers offshore AP from 64.4 to 64.2, whereas the consensus vote restores 64.8. The fusion operator, not the view count, therefore decides whether aggressive TTA helps or harms.
- A self-correcting inference cascade of two small correctors that write disjoint fields: a boundary-patch refiner confined to a ± 4-pixel band around predicted boundaries, which edits mask pixels only, and a 14-feature gradient-boosted consensus re-ranker, which edits detection scores only. Both are fitted on the frozen base model's own errors on the training split, draw no new annotation, and are audited against ground-truth leakage.
- A pretraining-regime study with full-recipe results. Under our 55-epoch protocol, COCO initialization moves the unmodified base to 27.7 inshore AP, 9.8 above the reported from-scratch-head result, and a controlled seven-direction sweep finds every architecture- and training-side modification flat, negative, or absorbed. The complete recipe reaches 44.5 inshore AP and 48.3 inshore AP_{75} on HRSID, and its training-free stack and refiner lift six further carrier–dataset pairs across five distinct architectures.

The rest of this article is organized as follows. Section II reviews related work. Section III details the proposed fusion operator and the two correctors. Section IV presents experiments on HRSID and PSeg-SSDD. Section V discusses the preservation regularity and the limitations of the study, and Section VI concludes this article.

II. Related Work

A. SAR Ship Instance Segmentation

Published SAR ship instance segmentation methods are predominantly specialized architectures trained from scratch, that is, an ImageNet-pretrained backbone with randomly initialized detection and mask heads, under long schedules on HRSID [1] and PSeg-SSDD [11]. LFG-Net exploits low-level feature guidance with super-resolution denoising to counter the scattering and noise characteristics of near-shore scenes [5]; a bounding-box-free Gaussian-heatmap formulation with centroid- and orientation-constrained losses reports 42.5 inshore mask AP on HRSID [4]; and DiffSARShipInst couples a diffusion-based instance decoder with aspect-ratio-matched asymmetric-convolution mask refinement and reports 42.6 inshore [3]. O2Former holds the strongest split-matched inshore result to date, 43.1 mask AP, by augmenting Mask2Former [8] with an optimized query generator and an orientation-aware polar-coordinate module under a 500-epoch from-scratch protocol in which its own Mask2Former baseline reaches only 17.9 inshore [2]. SARSAM adapts the Segment Anything Model by parameter-efficient tuning but reports only AP at IoU 0.50, which precludes full-range COCO-AP comparison [6].

This line of work varies the architecture under a fixed from-scratch protocol. Multi-stage pretraining has been studied for box-level SAR detection on SARDet-100K [7]. No SAR ship instance segmentation method isolates the pretraining regime as a variable, however, and none studies how the outputs of a trained model should be aggregated at inference. Under our protocol, COCO initialization moves the same Mask2Former from the reported 17.9 to 27.7 inshore mask AP, and we study recovery on that strong base instead of a new architecture.

B. Test-Time Augmentation, Weight Ensembling, and Inference-Side Recovery

Weight averaging is standard practice in detection: exponential moving average is a routine training configuration in modern detectors [12], with reported gains of up to +2.5 AP in from-scratch training but only +0.2 to +0.6 when fine-tuning strong checkpoints [13]; model soups average the weights of several models fine-tuned from a shared initialization [9]; and a recent empirical study of weight averaging concentrates its experiments on from-scratch regimes [14]. Multi-scale flip test-time augmentation (TTA) is equally common practice, and the rule that merges its outputs has been studied in box space. Soft-NMS decays overlapping scores instead of deleting them [15], weighted boxes fusion averages the coordinates of clustered boxes [16], and a learned aggregator weights augmented predictions for image classification [17]. Mask Scoring R-CNN [18] and IoU-Net [19] learn a mask- or box-IoU head inside the detector and rescore with it; our re-ranker regresses the same mask-IoU target outside the frozen model from cross-view consensus features. On the output side, BPR crops patches along predicted instance contours and re-segments them with a small high-resolution network as a model-agnostic post-processing stage [10]. PointRend [20] trains a point-wise head jointly with the segmentation network,

and SegFix [21] learns boundary offsets from the image and ground truth alone. BPR is adopted because it is the only one of the three fitted on a frozen model's own coarse masks. DiffusionDet makes the proposal count and the number of denoising steps an evaluation-time dial on an already-trained model, 45.8 to 46.8 box AP on COCO [22].

Two questions remain open in this family. The weight-averaging evidence is thinnest exactly where fine-tuning starts from a strong checkpoint. Beyond box-space NMS and weighted boxes fusion, no study asks which operator should merge the dozens of overlapping mask-level hypotheses that wide TTA produces. Section IV shows that the merge rule, not the view count, sets the outcome of aggressive TTA.

C. Adaptation and Priors on Strong Bases

A growing adapter literature argues that strong pretrained representations should be preserved, not overwritten. Mona shows that a multi-cognitive convolutional adapter surpasses full fine-tuning on COCO instance segmentation by +0.9 mask AP, whereas the linear low-rank adapter LoRand falls 0.4 AP below full fine-tuning (44.7 versus 45.1) [23], so both preservation and the adapter's inductive content matter; Frozen-DETR enriches a detector with tokens from a frozen vision foundation model and gains +2.9 AP on natural-image benchmarks [24]; and LayerScale's small-initialized per-channel residual scaling informs how such adapters are initialized [25]. A parallel line injects anisotropic priors for elongated targets: SegNeXt employs multiplicative strip attention [26], InceptionNeXt uses parallel strip kernels with a band-kernel sweet spot near $k = 11$ [27], Strip R-CNN applies sequential $1 \times k/k \times 1$ strip convolutions to remote-sensing detection [28], and in SAR the orientation-aware module of O2Former encodes anisotropy directly [2].

Preservation, however, proves necessary but not sufficient: on a COCO-pretrained Mask2Former, none of seven controlled architecture- and training-side arms paid, and the strip prior transferred at -0.9 inshore versus +0.6 offshore mask AP (Section IV). A plausible prior fails under regime shift, so our cascade recovers accuracy without modifying the base model at all.

III. Method

Section III-A gives an overview of the two-stage design. Section III-B details the consensus-gated mask-IoU vote fusion operator and the two carriers it rides on, and Sections III-C and III-D describe the two trained self-correctors.

A. Overview

Given a single-look SAR amplitude chip $\mathbf{x}$, in which the grayscale channel is replicated to three channels, ship instance segmentation asks for a set of scored binary masks $\mathcal{Y} = \{(\mathbf{m}_i, s_i)\}_{i=1}^{N_{\text{out}}}$, where $\mathbf{m}_i \in \{0,1\}^{800\times800}$ is the binary mask of the ith instance, $s_i \in [0,1]$ its confidence, and $N_{\text{out}} \le 100$ after score ranking under the COCO convention. Evaluation is COCO mask AP ($\times100$ throughout) on the HRSID official test split and on its inshore and offshore partitions, following the protocol of [2]. Throughout, θ denotes the full parameter vector of the segmentation network Φ_θ, and IoU abbreviates intersection over union.

The method has two stages, and Fig. 1 summarizes both. Stage A is a standard, unmodified Mask2Former-R50 [8] fine-tuned from COCO instance-segmentation initialization; its mask logits are predicted at stride 4 and bilinearly upsampled to the input resolution, a fact Section III-C returns to. Stage B updates no weight of Stage A and forms a cascade of three rungs at inference. The first rung is training-free, and its core is the first contribution: the consensus-gated mask-IoU vote fusion operator of Section III-B. It merges the candidate pool that two standard carriers produce, a weight soup and wide multi-scale flip test-time augmentation (TTA). The second and third rungs form the second contribution, two small trained correctors fitted on the frozen model's own training-split errors with no new annotation. These are a boundary-patch refiner that may write only mask pixels (Section III-C) and a consensus re-ranker that may write only scores (Section III-D). Every rung is optional and is reported separately in each table, and the training-free claim is scoped to the first rung.

The third contribution is a finding about the pretraining regime, not a module. The 9.8-point inshore gap between the from-scratch-head protocol of [2] and COCO initialization (Section I) motivates a preservation constraint: no layer, loss, or training schedule of Stage A is modified. Section IV-E4 supports the constraint with a seven-direction ablation in which none of the modifications earns a place in the recipe.

B. Consensus-Gated Mask-IoU Vote Fusion

The fusion operator consumes a pool of candidate masks, and two established practices, adopted here as carriers rather than claimed as contributions, produce that pool. The first is a weight soup [9]. Fine-tuning is stochastic: three independently seeded runs land within 0.1 inshore AP of one another yet err differently, and their checkpoints are averaged elementwise,

$$\theta^\star = \frac{1}{3} \sum_{s \in \{s_0,\, s_2,\, s_3\}} \theta_s, \tag{1}$$

where θ_{s_0}, θ_{s_2}, and θ_{s_3} are the parameter vectors of the three fine-tuned checkpoints. Two conditions are necessary for a usable average: all members are fine-tuned from the same COCO initialization with identical parameter keys, and their batch-normalization buffers are byte-equal. They are not sufficient: Section IV-E1, which sweeps the member count N from 1 to 6, finds seeds that meet both yet average well only within two groups. The soup is computed once, offline, yields a single deterministic checkpoint at $1\times$ inference cost, and lifts the three-seed inshore mean from 27.7 to 29.6 (+1.9). Three members were fixed by the training budget.

The second carrier is wide multi-scale flip TTA, common practice in detection that we deliberately push wider than usual: the souped model $\Phi_{\theta^\star}$ is run under the view set

$$\mathcal{V} = \{640,\, 800,\, 960,\, 1088\} \times \{\text{id},\, \text{hflip}\}, \qquad V = |\mathcal{V}| = 8, \tag{2}$$

where each pair $(\rho, f) \in \mathcal{V}$ resizes the chip to $\rho \times \rho$ and optionally applies a horizontal flip. The framework resamples

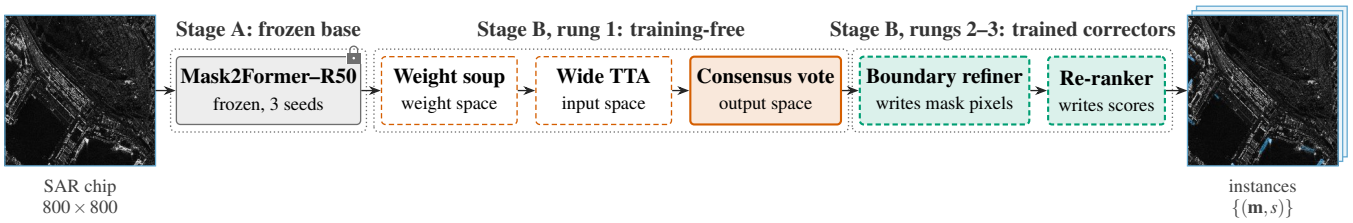


Fig. 1. One frozen base and three inference-time rungs, each writing a different output field. Stage A is a COCO-initialized Mask2Former-R50, never updated (padlock; contribution 3). Rung 1 of Stage B is training-free: soup and eight-view TTA carry the consensus-vote operator (contribution 1); rungs 2–3 are trained correctors on disjoint fields (contribution 2). Color key: gray with padlock, frozen; vermillion, rung-1 operators; green, dashed, correctors; blue, data. Both rasters are real tensors of one HRSID chip. AP per rung: Table IV.

every view's masks to the native 800×800 resolution and inverts the flip, so the pooled set holds $K \le 8 \times 100 = 800$ scored candidates per image. Running the frozen network at 640 to 1088 pixels needs no retraining: the backbone and pixel decoder are fully convolutional, and the 100 learned queries carry no notion of resolution. Fine-tuning already rescales every training chip by a random factor in $[0.5, 2.0]$ before the 800×800 crop, so all four test scales lie inside the fitted range. A six-view core $\{736, 800, 864\} \times \{\mathrm{id}, \mathrm{hflip}\}$ already lifts single-checkpoint inshore AP by +7.0 on the three-seed mean, and widening to the eight views of Eq. (2) adds a further +2.0 inshore on seed s_0, because the upsampled scales place more pixels on pixel-starved near-shore hulls. The wider set is not free under the default box-NMS merge: on the soup, going from six to eight views moves offshore AP from 64.4 to 64.2 (Section IV-E1). We push the views this wide only because the gated operator below decides, cluster by cluster, where averaging is safe. What the pooled views demand is a fusion rule.

Merging up to 800 overlapping candidates is where aggressive TTA either pays or collapses, and three failure modes are observed on this pool. Box-space non-maximum suppression (NMS) deletes berthed neighbors whose boxes overlap although their masks do not. Score-decay suppression (matrix-NMS) leaves the roughly eight near-duplicates per object in the ranking. Unconditional pixelwise voting erodes well-resolved offshore boundaries below the no-TTA base. Section IV-E2 quantifies all three on the identical candidate set. The fuser, not the views, is the safety condition, so the operator clusters in mask-IoU space, deletes duplicates hard, and averages only where the ensemble itself signals disagreement.

All K candidate masks are flattened into rows of $\mathbf{M} \in \{0,1\}^{K \times HW}$, and the pairwise mask-IoU (Jaccard) matrix is

$$J_{jk} = \frac{\mathbf{m}_j^\top \mathbf{m}_k}{\mathbf{1}^\top \mathbf{m}_j + \mathbf{1}^\top \mathbf{m}_k - \mathbf{m}_j^\top \mathbf{m}_k}, \tag{3}$$

where $\mathbf{m}_j^\top \mathbf{m}_k$, the (j,k) entry of the Gram matrix $\mathbf{M}\mathbf{M}^\top$, counts intersection pixels and $\mathbf{1}^\top \mathbf{m}_j$ is the area of mask j. Candidates are consumed greedily in score order. With $\mathcal{U}$ the set of not-yet-consumed indices, one cluster is formed per iteration as

$$\mathcal{C} = \{r\} \cup \{j \in \mathcal{U} : c_j = c_r,\ J_{rj} \ge \tau\}, \qquad r = \arg\max_{k \in \mathcal{U}} s_k, \tag{4}$$

where r indexes the seed (the highest-scoring unconsumed candidate), c_j is the class label of candidate j, and $\tau = 0.5$ is the mask-IoU threshold. Each cluster is read out under a cross-view agreement gate,

$$\mathbf{m}^{\mathrm{out}}(p) = \begin{cases} \mathbf{1}\Big[\sum_{j \in \mathcal{C}} \mathbf{m}_j(p) \ge \frac{|\mathcal{C}|}{2}\Big], & 2 \le |\mathcal{C}| \le S_{\max}, \\ \mathbf{m}_r(p), & \text{otherwise}, \end{cases} \tag{5}$$

where p ranges over pixels and $S_{\max} = 6$ bounds the agreement band. The output score is

$$s^{\mathrm{out}} = s_r \left(|\mathcal{C}|/V\right)^{\gamma}, \qquad \gamma = 0, \tag{6}$$

where $\gamma = 0$ reduces the rule to plain seed-score inheritance; the general form is kept because the consensus ratio $|\mathcal{C}|/V$ is the quantity the gate consumes, and Section III-D returns to it. The output box is regenerated from the fused mask. Algorithm 1 lists the complete procedure, and Fig. 2 shows the operator on a real candidate pool, together with the two failure modes that motivate its gate.

Figure 2 makes the vote concrete. One vote is one mask that one view assigned to a ship, and votes are cast pixel by pixel. A pixel enters the fused mask when at least half of the member masks contain it. The 16 highest-scoring candidates of the chip form four clusters of sizes 6, 7, 1, and 2. In the six-member cluster a pixel survives with three or more votes. Pixels that only one or two views claim are dropped, and the hull on which the views agree is kept. In the two-member cluster one vote suffices, and the output is the union of the two masks. The seven-member cluster exceeds $S_{\max}$. Averaging seven masks that are each correct but offset by one to two pixels would only trim its bow and stern, so its seed mask is emitted verbatim. The singleton passes through unchanged, and in every case the seed score is inherited under $\gamma = 0$.

The agreement band reads the ensemble's own cross-view consensus as a per-instance variance estimate. Clusters above the band are clean detections whose seed masks any averaging can only erode, and singletons have nothing to vote over. Only in-band clusters, typically ambiguous inshore ships, are denoised by the pixel vote. Composed, the training-free rung

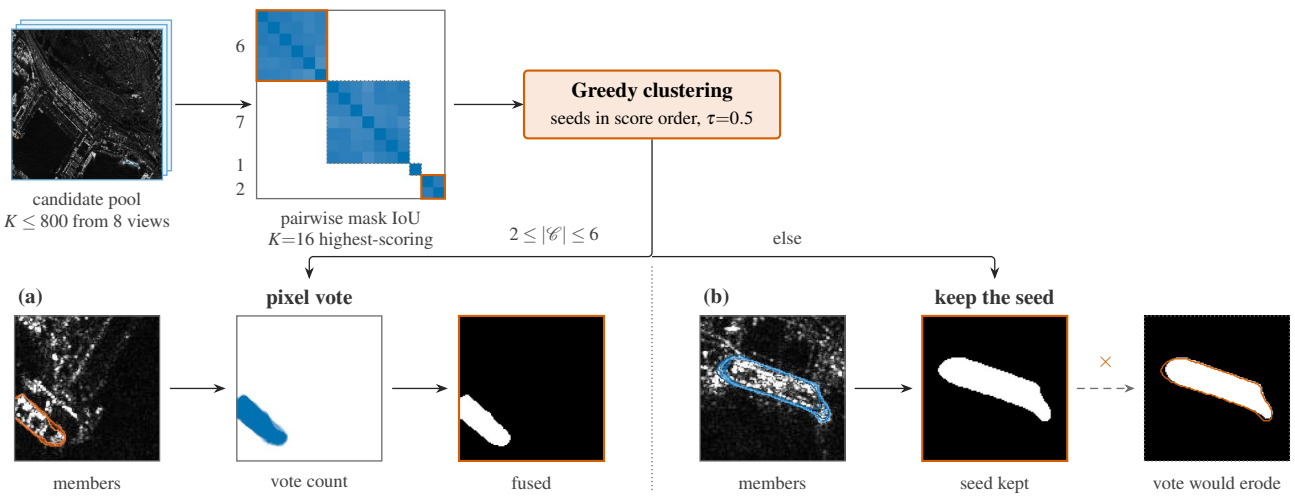


Fig. 2. Consensus-gated vote fusion on a real HRSID inshore candidate pool. Top: measured pairwise mask IoU of 16 candidates in cluster order (white, IoU 0; blue, IoU 1); solid vermillion marks the two in-band clusters, dotted gray the two out-of-band. (a) $|\mathcal{C}|$=6: members, vote count, fused mask. (b) $|\mathcal{C}|$=7: the seed is kept; the vote not taken, inside the vermillion seed contour, trims the hull. Both crops are 19.25 mm wide (118 and 110 px). Operators and band: Table VI.

Algorithm 1 Consensus-Gated Mask-IoU Vote Fusion

Require: candidates $\{(\mathbf{m}_k, s_k)\}_{k=1}^{K}$ from $V = 8$ views; mask-IoU threshold $\tau = 0.5$; agreement band $[2, S_{\max}]$ with $S_{\max} = 6$; $\gamma = 0$
Ensure: fused instance set $\mathcal{Y}$

sort candidates by score, descending; mark all unused; $\mathcal{Y} \leftarrow \emptyset$
while unused candidates remain **do**
  $r \leftarrow$ highest-scoring unused candidate; $\mathcal{C} \leftarrow \{r\}$
  absorb every unused same-class j with $J_{rj} \geq \tau$ into $\mathcal{C}$
  mark $\mathcal{C}$ used
  if $2 \leq |\mathcal{C}| \leq S_{\max}$ **then**
    $\mathbf{m}^{\text{out}} \leftarrow$ pixels held by at least half of $\mathcal{C}$ {denoise}
  else
    $\mathbf{m}^{\text{out}} \leftarrow \mathbf{m}_r$ {singleton or near-unanimous: preserve seed verbatim}
  end if
  $s^{\text{out}} \leftarrow s_r\,(|\mathcal{C}|/V)^{\gamma}$; box $\leftarrow$ `mask2bbox`$(\mathbf{m}^{\text{out}})$
  append $(\mathbf{m}^{\text{out}}, s^{\text{out}})$ to $\mathcal{Y}$
end while
return top-100 of $\mathcal{Y}$ by score

lifts the seed-s_0 base (27.7/45.6/59.6) by +11.1/+8.2/+5.2 with no weight updated, of which +1.7/+1.2/+0.6 is the fusion operator over box-NMS on the same candidates. Section IV-E2 and Table VI report the operator comparison and the band-ceiling sweep.

C. Boundary-Patch Refinement

The training-free rung recovers detection and recall, not boundary precision. Mask logits predicted at stride 4 are upsampled $4\times$ bilinearly, so predicted contours carry a soft halo of roughly one to two pixels, and on a typical 9×18-pixel inshore hull a one-pixel boundary shift already costs about 0.15–0.25 IoU; no training-free control we tested moves this residual (Section IV). The second rung is therefore a trained corrector in the boundary-patch-refinement (BPR) family [10], constructed so that it cannot alter anything except boundary pixels; Fig. 3 shows its training and application paths.

The refiner is a separately trained network, distinct from the frozen base, and its supervision comes from that base's own mistakes. The frozen souped base is run single-view, without TTA, on the training split. The halo it targets originates in the stride-4 mask head and is present in every view, so the cheaper single-view dump is used. Predictions with score ≥ 0.2 are matched to the training ground truth at mask IoU ≥ 0.5, which pairs 10 297 instances on 3641 HRSID training images. Along each matched contour, 64×64 windows centered on boundary pixels, at most one per 32×32 grid cell, are cropped from the SAR chip, the predicted mask, and the ground-truth mask. This yields 37 009 patch triplets, 1850 of which are held out to select the checkpoint. The refiner is a 2.35M-parameter UNet that reads two channels, the grayscale patch and the coarse mask, and predicts the refined mask. It is trained from scratch for 20 epochs on each dataset with a combined binary cross-entropy and Dice objective [10]. The remaining training details are given in Section IV-B. At test time the refiner re-segments the boundary patches of each detected instance, and the refined prediction is written back only inside a ±4-pixel band around the original boundary,

$$\begin{aligned} \mathbf{m}^{\text{final}}(p) &= b(p)\,\hat{\mathbf{m}}(p) + \big(1 - b(p)\big)\,\mathbf{m}^{\text{out}}(p), \\ b(p) &= \mathbf{1}\big[d\big(p, \partial\mathbf{m}^{\text{out}}\big) \leq 4\big], \end{aligned} \tag{7}$$

where $\hat{\mathbf{m}}$ is the refiner output stitched over the patches, $\partial\mathbf{m}^{\text{out}}$ the boundary of the fused mask of Eq. (5), and $d(\cdot,\cdot)$ the pixel distance to that boundary; overlapping patch logits are averaged inside the band and thresholded at 0.5.

The guard is what makes the rung safe. Every pixel outside the band is bit-identical to the preceding rung's output, and detection scores, ranks, and instance identities are untouched, so the refiner can displace boundaries by a few pixels but can neither add, remove, nor re-rank detections. Section IV quantifies the effect and audits the application path for leakage.

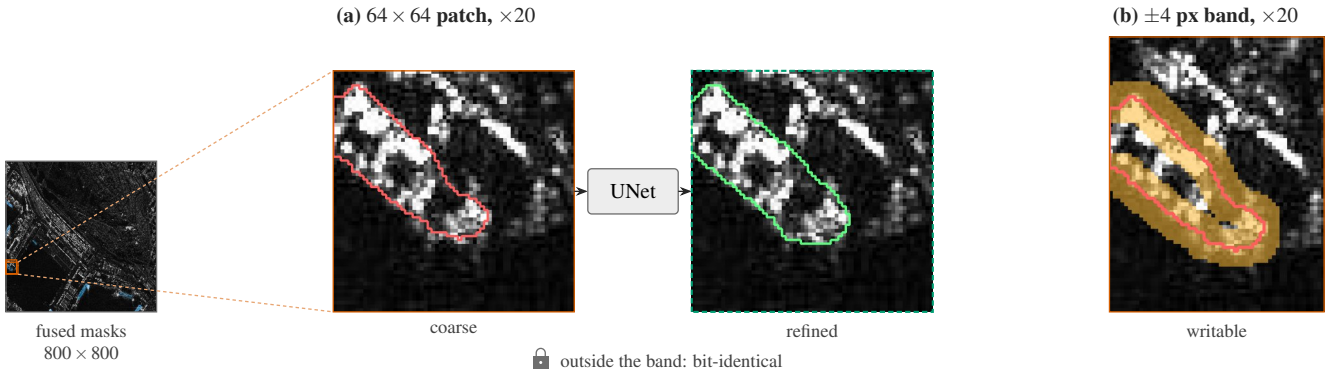


Fig. 3. Boundary-patch refinement on the real fused masks of one HRSID inshore chip. The two boxes are the windows the dump recorded; both panels are ×20 of the overview, letters top left. (a) A real 64 × 64 patch before and after the 2.35M-parameter UNet refiner (red: coarse; green: refined). (b) The real ±4-pixel annulus $\mathrm{dilate}(\mathbf{m}, 4) \ominus \mathrm{erode}(\mathbf{m}, 4)$; amber is writable, red the fused hull. Outside the band every pixel is bit-identical; scores and ranks are untouched. Effect on AP: Table IV.

D. Learned Consensus Re-Ranking

The third rung treats the ranking axis of the residual. Mask AP is a ranking metric, so a candidate whose mask is accurate but whose score falls below a worse neighbor forfeits precision at every recall level it should have dominated. The fusion operator of Section III-B already computes the quantity that predicts mask quality, the cross-view consensus of each cluster, and then discards it: Eq. (6) reduces to seed-score inheritance at $\gamma = 0$ because no fixed exponent on $|\mathcal{C}|/V$ improved all three splits at once. Rather than search harder for a formula, we stop writing one. The base model also errs on the training split, and those errors are free supervision, so the map from consensus evidence to mask quality is fitted instead of guessed.

Each fused instance is described by a 14-dimensional vector of scalars,

$$\begin{aligned} \boldsymbol{\phi} &= \left[\boldsymbol{\phi}_{\mathrm{c}}^{\top},\ s^{\mathrm{out}},\ \boldsymbol{\phi}_{\mathrm{g}}^{\top}\right]^{\top} \in \mathbb{R}^{14}, \\ \boldsymbol{\phi}_{\mathrm{c}} &= \left[\,|\mathcal{C}|,\ n_v,\ \bar{J},\ J_{\min},\ s_r,\ \Delta s\,\right]^{\top}, \\ \boldsymbol{\phi}_{\mathrm{g}} &= \left[\log(1{+}a),\ \log(1{+}w),\ \log(1{+}h),\right. \\ &\qquad \left. w/h,\ a/(wh),\ c_x,\ c_y\,\right]^{\top}, \end{aligned} \tag{8}$$

where $\boldsymbol{\phi}_{\mathrm{c}}$ collects the consensus statistics of the cluster. Its first two entries are the cluster size $|\mathcal{C}|$ of Eq. (4) and the number n_v of distinct views of Eq. (2) contributing to it. Next come the mean $\bar{J}$ and minimum $J_{\min}$ pairwise mask IoU of Eq. (3) inside the cluster. The seed score s_r and the spread Δs between the largest and smallest member score complete the block. The scalar s^{out} is the score the fusion stage emits. The block $\boldsymbol{\phi}_{\mathrm{g}}$ holds geometry: the mask area a, the width w and height h of the mask box, and the box center (c_x, c_y) normalized by image width and height. Under the deployed setting $\gamma = 0$ the two score channels coincide, which costs a tree ensemble nothing and keeps the same extractor valid for $\gamma > 0$. No image pixel is read anywhere in Eq. (8): every entry is a by-product of computation the stack has already performed, which is what makes the operator cost microseconds per instance.

The regression target is the mask IoU between the candidate and its best-matching training annotation, and zero when the candidate overlaps none; every candidate above a low score floor contributes a row, kept and rejected alike, so the model sees the failures and not only the survivors. A histogram-based gradient-boosted regression tree g is fitted to these rows on the training split alone, and its prediction is blended into the detection score as

$$s^{\mathrm{final}} = \alpha\, \mathrm{clip}\big(g(\boldsymbol{\phi}), 0, 1\big) + (1 - \alpha)\, s^{\mathrm{out}}, \tag{9}$$

where $\alpha \in [0, 1]$ trades the learned quality estimate against the network's own confidence and the clip keeps the regressor inside the score range. The mixing weight is the only hyper-parameter and is selected on held-out training images alone, never on test data; the fitting protocol and its cost, seconds of CPU time, are given in Section IV.

Two safety properties follow from what the operator is allowed to write. First, masks, instance identities, and the size of the output set are bit-identical to the input: only the score field changes, so re-ranking can neither create, delete, nor reshape a detection, and a detection that matches no fusion-stage candidate keeps its original score untouched. Second, the refiner and the re-ranker write to disjoint fields, mask pixels inside a ±4-pixel band versus scores, so neither can undo the other; we apply the refiner first so that the reported masks are the refined ones. A carrier or a dataset new to the recipe needs its own g, since the consensus statistics belong to that carrier, and refitting one costs under a CPU-minute and no GPU time.

IV. Experiments

Sections IV-A–IV-C fix the experimental protocol, and Section IV-D compares with the state of the art. Section IV-E ablates the contributions in order. Section IV-F transfers the recipe to PSeg-SSDD and six further carrier–dataset pairs, and Section IV-G records the directions that did not help.

A. Dataset

HRSID [1] comprises 5604 SAR chips of 800×800 pixels with instance-level mask annotations, split into 3642 training and 1962 test images. Following the inshore/offshore protocol of prior work [2], the official test set is additionally partitioned into 369 inshore and 1593 offshore images. We report all three

splits throughout; the inshore split is where the open problem lives. Cross-dataset experiments use PSeg-SSDD, with 928 training and 232 test images at 512×512; its test set divides into 46 inshore and 186 offshore scenes under the same protocol [2].

B. Implementation Details

The base model, denoted c0, is a standard Mask2Former (M2F) [8] with a ResNet-50 backbone and 100 queries, initialized from COCO instance-segmentation weights; an auxiliary mask-IoU rescoring head is disabled ($\lambda = 0$), its gain being subsumed by the soup (Section IV-E4). On HRSID the base is fine-tuned at 800×800 with AdamW at an initial learning rate of 2.5×10^{-5} with step decay, batch size 2 with 2-step gradient accumulation (effective batch 4), for 100 872 iterations, i.e., 55 epochs over the 3642-image training split; on PSeg-SSDD the same protocol, at batch 4 without accumulation (the same effective batch), runs for 8352 iterations, i.e., 36 epochs over the 928-image split, all on a single NVIDIA RTX 3090 Ti. The wide-TTA view set on HRSID is $\{640, 800, 960, 1088\} \times \{\text{identity}, \text{flip}\}$; on PSeg-SSDD the scales rescale to $\{416, 512, 608, 704\}$ in proportion to the smaller chips, and the consensus band $2 \leq |\mathcal{C}| \leq S_{\max} = 6$ is carried over unchanged.

The boundary-patch refiner of Section III-C is trained on square boundary patches of side 64, upscaled $2\times$; its 20 epochs take about 19 minutes on HRSID. The re-ranker is a histogram-based gradient-boosted regression tree (300 boosting iterations, learning rate 0.08, least-squares loss) fitted on 51 781 candidate rows from the 3642 HRSID training images (5122 rows from 928 on PSeg-SSDD) in 19 s (9 s) on CPU. Its blend weight α is swept over $\{0.3, 0.5, 1.0\}$ on a 4:1 image-identifier partition of the training split, landing at $\alpha = 0.5$ on HRSID and $\alpha = 1.0$ on PSeg-SSDD, after which the regressor is refitted on all rows; on the held-out training images its prediction tracks the realized mask IoU at a Spearman rank correlation of 0.68 on HRSID and 0.86 on PSeg-SSDD. The measured end-to-end cost of the full method is approximately $8\times$ a single-view pass. Within it, fusion adds one Gram product over $K \leq 800$ binary masks, the refiner a per-instance boundary-patch pass, and the re-ranker one tree evaluation per instance. Flip TTA on the Cascade R-CNN carrier of Section IV-F required a parameter-free repair of an mmdetection mask-head merge routine, output-identical to the stock head on non-flipped pipelines.

C. Evaluation Setting

Every number in this section is COCO mask AP $\times 100$, reported to one decimal, on the official, inshore, and offshore splits. The base was trained with three independent seeds (s0/s2/s3). Across them the single-model base scores 27.7±0.0 inshore, 45.7±0.2 official, and 59.8±0.3 offshore (mean±sample s.d.). Single-seed differences within this 0.3-AP band are treated as noise. The ladders print the seed-s_0 checkpoint, 27.7/45.6/59.6, as their base row and quote increments over it. Inference is deterministic: the refiner re-run reproduces all reported figures to four decimals and the re-ranker regenerates its outputs byte-identically, so the cascade adds no stochastic stage of its own. The refiner's training seed is the one remaining source of variation, worth at most 0.5 point (Section IV-E3). Every Mask2Former configuration of ours is evaluated at its final training iteration; the validation hook monitors the official test split for logging only and selects no checkpoint. The exceptions are marked where they occur: the 200-epoch control of Section IV-E4, and the competitor rows of Section IV-D2, which also serve as the PSeg-SSDD carriers of Section IV-F2. Where a confidence interval is quoted, it comes from a paired percentile bootstrap over test images. Each of 1000 resamples draws images with replacement and both configurations are re-scored on the same resample; the 2.5th and 97.5th percentiles of the paired AP difference form the interval. The provenance of every competitor row is stated in the caption and footnote of its table.

Hyperparameter selection. The settings of the training-free rung, namely the mask-IoU threshold $\tau = 0.5$, the band ceiling $S_{\max} = 6$, the exponent $\gamma = 0$, the eight-view scale set, and the choice of fusion operator (Section IV-E2), were fixed by sensitivity sweeps on the three HRSID test splits; no held-out split was used. Two facts bound the optimism this introduces. First, the sweep surface is flat: $S_{\max} = 4/5/6$ gives inshore 38.4/38.6/38.8 with offshore unchanged at 64.8, a 0.4-point spread. Second, the settings were then frozen and carried to PSeg-SSDD with only the view scales rescaled to the chip size, and to six further carriers with no retuning, and every one of those endpoints improved (Sections IV-F1 and IV-F2). The re-ranker's blend weight, the one swept setting of the learned tiers, is chosen on a held-out partition of the training split (Section IV-B), not on test images. A training-side re-sweep could not have improved the choice. On a fixed random 20% of the HRSID training images (728 chips the model had seen), the souped eight-view stack scores 57.6 for every $S_{\max} \in \{4, 5, 6, 7\}$, and $\tau = 0.4/0.5/0.6$ gives 57.7/57.6/57.4. On images it was fitted to, the model seldom disagrees with itself across views, so the band ceiling never engages there; only the test splits expose the small differences.

D. Comparison With State-of-the-Art Methods

1) Comparison With Published Methods on HRSID: Table I compares against the reported HRSID inshore and offshore results under the same split protocol; competitor rows are as reported, sources in the footnote. The offshore AP_{50} of 97.8 and AP_{75} of 90.0 relayed for DiffSARShipInst [3] have, to our knowledge, not been independently reproduced.

With both correctors, inshore reaches 44.5 AP, 48.3 AP_{75}, and 42.0 AP_s, +1.4, +2.4, and +2.1 over O2Former, the strongest reported method; the refiner tier alone already leads these columns. The training-free stack posts 75.6 inshore AP_{50} against O2Former's 69.5, but its inshore AP of 38.8 and AP_{75} of 36.1 trail their 43.1 and 45.9: it recovers detections, not boundaries. Offshore, the cascade trails the two strongest specialized methods (70.1 against 71.7 and 70.9), although its AP_{50} of 98.6 is the highest in the table. No competitor reports the official split; Section IV-E1 gives our figures there.

One caveat dominates: the two row groups are not at the same operating point. The competitor protocol trains segmen-

TABLE I
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE HRSID INSHORE AND OFFSHORE SPLITS (MASK AP $\times 100$). COMPETITOR ROWS AS REPORTED; "SCRATCH" DENOTES AN IMAGENET BACKBONE WITH A FROM-SCRATCH SEGMENTATION HEAD; "N/R" DENOTES NOT REPORTED

				Inshore				Offshore			
Method	Init	Epochs	Infer.	AP	AP_{50}	AP_{75}	AP_s	AP	AP_{50}	AP_{75}	AP_s
Full method + BPR + re-ranking (ours)	COCO	3×55	$\sim 8\times$	**44.5**	76.7	**48.3**	**42.0**	70.1	**98.6**	88.0	69.4
Full method + BPR (ours)	COCO	3×55	$\sim 8\times$	43.6	76.3	47.0	41.4	69.1	**98.6**	87.4	68.7
O2Former† [2]	scratch	500	$1\times$	43.1	69.5	45.9	39.9	**71.7**	97.5	**91.0**	**71.1**
DiffSARShipInst‡ [3]	n/r	n/r	$1\times$	42.6	76.1	44.1	40.3	70.9	97.8	90.0	70.3
Gaussian-Heatmap§ [4]	n/r	n/r	$1\times$	42.5	**79.1**	39.9	41.0	68.0	97.7	85.1	67.4
RTMDet† [29]	scratch	500	$1\times$	40.4	75.9	38.7	39.3	67.4	97.7	83.7	66.8
HQ-ISNet† [30]	scratch	500	$1\times$	39.6	72.5	41.0	39.0	67.9	97.5	87.1	66.5
Full method, training-free (ours)	COCO	3×55	$\sim 8\times$	38.8	75.6	36.1	35.8	64.8	98.5	81.7	64.1
Cascade Mask R-CNN† [31]	scratch	500	$1\times$	35.6	59.1	40.1	32.6	67.4	96.8	85.7	66.7
Mask R-CNN† [32]	scratch	500	$1\times$	34.0	56.5	38.3	31.0	65.3	95.8	83.4	64.5
c0 base, seed s_0 (ours)	COCO	55	$1\times$	27.7	61.8	22.1	24.3	59.6	97.2	71.2	58.6
Mask2Former† [8] (their baseline)	scratch	500	$1\times$	17.9	36.9	16.2	14.7	60.6	95.2	74.9	59.5

Bold denotes the best value in each column; ties at one decimal are both marked. All methods use a ResNet-50 backbone where reported. Epochs: training budget, 3×55 denoting three 55-epoch seeds averaged into one weight soup. Infer.: inference cost relative to one single-view pass; competitor rows report single-view inference or no test-time augmentation and are entered as $1\times$. †As tabulated by O2Former [2], Table 3; not re-evaluated by us. ‡Primary source paywalled; figures relayed via O2Former's table, cross-checked against the article abstract. §From [4], Table II; protocol not reported.

tation heads from scratch for 500 epochs on an ImageNet backbone; we fine-tune a COCO-initialized model for 55 epochs. The same Mask2Former scores 17.9 inshore under their protocol and 27.7 under ours, a 9.8-point separation before any contribution of this article (Section IV-E4). The defensible claims are the same-base recovery of Section IV-E1 and the two audited correctors stacked on it.

2) Comparison Under One Protocol on PSeg-SSDD: PSeg-SSDD is small enough to retrain competitors on our own hardware, so every row of Table II is ours: six methods spanning four families under one protocol. The four generic architectures cluster between 38.9 and 41.2 inshore, whereas the two SAR-specific modules jump to 52.9 and 53.2, so on this dataset the near-shore regime, not the choice of segmentation head, sets the level. One selection detail is disclosed, not corrected: every competitor row is its best-on-test checkpoint (Epochs column), whereas our rows are the final iteration. The two reimplemented rows were selected on the inshore split they lead, so their inshore entries are upper bounds. Our unmodified soup base already reaches 51.9 inshore with no SAR-specific component, the training-free stack passes both specialized rows on all three splits at 54.0/63.3/66.9, and with both correctors the margins are +8.9 inshore over the strongest competitor, +6.7 official, and +4.7 offshore over Mask R-CNN. The recipe also lifts the OAEM carrier itself: the stack takes it from 53.2/60.8/64.0 to 54.3/60.6/63.3 and the refiner to 59.9/64.2/66.1. That is a composed +6.7 inshore, with inshore AP_{75} rising from 55.9 to 74.9. The SAR-specific module thus neither blocks the recipe nor is needed by it: the unmodified soup base with the same two rungs stays ahead at 60.0/66.3/68.8. The stack's small official and offshore losses on this carrier (-0.2 and -0.7) are the erosion of already stable predictions that Section IV-E2 describes.

A detection-level re-reading with greedy score-descending matching at mask IoU ≥ 0.5 (Table III) confirms that the margin is not an artifact of the AP integral. At a fixed 0.30 score threshold the refined method detects 169 of the 172 inshore ships ($P_d = 0.983$) against 159 ($P_d = 0.924$) for the better specialized row, and it reads $F_1^* = 0.902$ against 0.882 at each method's own optimal threshold. Offshore is saturated, all three detecting 369 of the 374 ships. In the precision-recall curves of Fig. 4, the refined pipeline is the only method of the eight that reaches full recall at nonzero precision at IoU $= 0.50$, where it reads 95.7 against a competitor best of 92.2, and at IoU $= 0.75$ it leads at 72.5 against a competitor best of 58.8: the training-free operators recover detections, and the boundary refiner converts them into strict-overlap accuracy.

3) Qualitative Comparison: Figure 5 puts the cascade and two detectors trained under our own protocol, Mask R-CNN and Cascade Mask R-CNN, on the same HRSID scenes. SOLOv2 [37], the weakest at 19.0 inshore (Section IV-F2), is omitted for width. Scene selection is programmatic and favors us by construction. Per inshore test image the stronger competitor is the one with fewer visible errors, counted as missed targets plus half its false alarms. Images are ranked by the reduction of that count to our final column, ties broken by the false-alarm difference, and the two top-ranked scenes are shown. The offshore scene is the error-free one with the largest inter-method mask-IoU spread. On the 95-ship chip of row (a) the training-free stack finds 66 ships against 23 and 28 for the competitors at comparable false-alarm counts. Across the two inshore rows it finds 1.9 to 2.9 times as many targets, the signature of recall recovery, not of a lowered threshold. In the offshore row (c), solved by every method, mask quality alone separates the competitors (0.782 and 0.777 mean matched IoU) from the stack at 0.828 and both correctors at 0.873. The competitor columns are trained at our budget, so this says nothing about the reported systems of Table I.

Figure 6 extends the comparison to the eight PSeg-SSDD methods of Table II, with scenes selected by the same rule, and the honest reading of it is narrower than the table margins. Exactly one of the 46 inshore test scenes yields a strictly positive count gain for our columns over the best competitor: the 13-ship quay of row (a), which both of our columns sweep at 13 of 13. Row (b) is a count tie that separates on matched IoU. Row (c) is a tie for all eight methods, with the refined

TABLE II
COMPARISON WITH SIX INDEPENDENTLY TRAINED METHODS ON PSEG-SSDD (MASK AP ×100). EVERY ROW, COMPETITORS INCLUDED, WAS TRAINED AND SCORED BY US UNDER ONE EVALUATOR. ROWS ARE ORDERED BY INSHORE AP

Method	Family	Init	Epochs	Inshore	Official	Offshore
YOLACT [33]	single-stage prototype	ImageNet	55 (best@31)	38.9	55.7	63.4
Mask R-CNN [32]	two-stage region-based	ImageNet	50 (best@41)	40.8	57.8	65.1
Mask R-CNN + DCN [34]	two-stage + deformable conv. (DCN)	ImageNet	50 (best@19)	41.0	57.8	64.6
CondInst [35]	single-stage dynamic kernel	COCO	36 (best@36)	41.2	54.3	60.1
M2F + DenoDet-style NDM (our reimpl.)* [36]	query-based + SAR module	COCO	36 (best@27)	52.9	60.5	63.7
M2F + OAEM (our reimpl.)* [2]	query-based + SAR module	COCO	36 (best@36)	53.2	60.8	64.0
+ stack (1 seed)	+ training-free stack	COCO	36 (best@36)	54.3	60.6	63.3
+ stack + BPR (1 seed)	+ boundary refiner	COCO	36 (best@36) +20 BPR	59.9	64.2	66.1
Soup base (ours)	query-based, unmodified	COCO	3 × 36	51.9	60.9	64.8
Full method (ours)	+ training-free stack	COCO	3 × 36	54.0	63.3	66.9
Full method + BPR (ours)	+ boundary refiner	COCO	3 × 36 +20 BPR	60.0	66.3	68.8
Full method + BPR + re-ranking (ours)	+ learned re-ranking	COCO	3 × 36 +20 BPR	**62.1**	**67.5**	**69.8**

Bold denotes the best value in each column. All rows were fine-tuned on the identical 928-image training split and scored with the identical COCO evaluator at 512 × 512; the ImageNet-initialized group received the larger epoch budget. Checkpoints: all competitor rows use their best-on-test checkpoint, the epoch in parentheses, and the two reimplemented rows were selected on the inshore split they lead; our rows are the final iteration, which favors the competitors. *No public code exists for either module, so these two rows are our own reimplementation of the described modules on our Mask2Former carrier; they measure that reimplementation, not the original systems. The two rows beneath the OAEM row apply our recipe to that same OAEM checkpoint: the training-free stack, then a refiner fitted on its own training-split predictions (8013 patches, 20 epochs); single seed, no soup. For reference, O2Former reports on this dataset, under its own 500-epoch from-scratch protocol, a Mask2Former baseline of 52.5 inshore and 67.2 offshore, and 55.2 and 71.0 with OAEM added [2]. Three further methods were run and excluded: SOLO diverged, PointRend and InstaBoost were trained at 1333 × 800 but tested at 512 × 512 (a resolution artifact), and FreqFusion collapsed in its own training log.

TABLE III
FALSE-ALARM / MISSED-DETECTION ACCOUNTING ON PSEG-SSDD (MASK IOU ≥ 0.5, GREEDY SCORE-DESCENDING MATCHING) AT A FIXED SCORE THRESHOLD OF 0.30 AND AT THE PER-METHOD, PER-SPLIT F_1-OPTIMAL THRESHOLD

	Threshold 0.30			F_1-optimal threshold			
Method	P_d	FA	F_1	P_d^*	FA*	F_1^*	thr*
Inshore split (46 images, 172 GT ships)							
YOLACT	0.709	1.957	0.722	0.692	0.587	0.748	0.40
Mask R-CNN	0.721	1.326	0.695	0.663	0.587	0.728	0.70
Mask R-CNN + DCN	0.738	1.739	0.670	0.663	0.717	0.715	0.65
CondInst	0.657	**0.217**	0.766	0.744	0.565	0.785	0.20
M2F + NDM*	0.924	1.000	**0.844**	0.843	**0.261**	0.881	0.85
M2F + OAEM*	0.907	1.109	0.823	0.866	0.370	0.882	0.80
Base (ours)	0.866	0.717	0.842	0.855	0.326	0.880	0.40
Full method (ours)	0.977	1.435	0.828	0.924	0.565	0.891	0.55
Full + BPR (ours)	**0.983**	1.413	0.833	**0.936**	0.522	**0.902**	0.55
Offshore split (186 images, 374 GT ships)							
YOLACT	0.960	0.075	0.961	**0.992**	0.118	0.967	0.15
Mask R-CNN	0.933	0.081	0.946	0.928	0.048	0.951	0.75
Mask R-CNN + DCN	0.933	0.091	0.943	0.922	**0.038**	0.950	0.75
CondInst	0.928	**0.038**	0.953	0.976	0.081	0.968	0.10
M2F + NDM*	**0.987**	0.075	0.975	0.984	**0.038**	**0.983**	0.80
M2F + OAEM*	**0.987**	0.054	**0.980**	0.984	**0.038**	**0.983**	0.85
Base (ours)	0.963	0.167	0.941	0.936	0.075	0.949	0.40
Full method (ours)	**0.987**	0.086	0.972	0.979	**0.038**	0.980	0.50
Full + BPR (ours)	**0.987**	0.086	0.972	0.979	**0.038**	0.980	0.50

$P_d = \mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$, so the missed-detection rate is $1 - P_d$; FA: false alarms per image; $F_1 = 2\mathrm{TP}/(2\mathrm{TP}+\mathrm{FP}+\mathrm{FN})$; thr*: the F_1-optimal threshold, swept over 0.05 : 0.05 : 0.90. Bold marks the best value in each column (P_d and F_1: higher is better; FA: lower is better); ties are both marked. All rows are our own runs under one identical protocol. *Our reimplementation on our carrier, not the original system (Table II). P_d at a fixed threshold can be inflated by false alarms; read it together with FA, or use the F_1 columns.

column leading on mean matched IoU at 0.898 against a competitor best of 0.888. On the easier dataset the specialized baselines sit close to detection saturation, so the margin the recipe carries there lives at the mask level and in the ranking, exactly where Sections IV-E3 and IV-F1 put it.

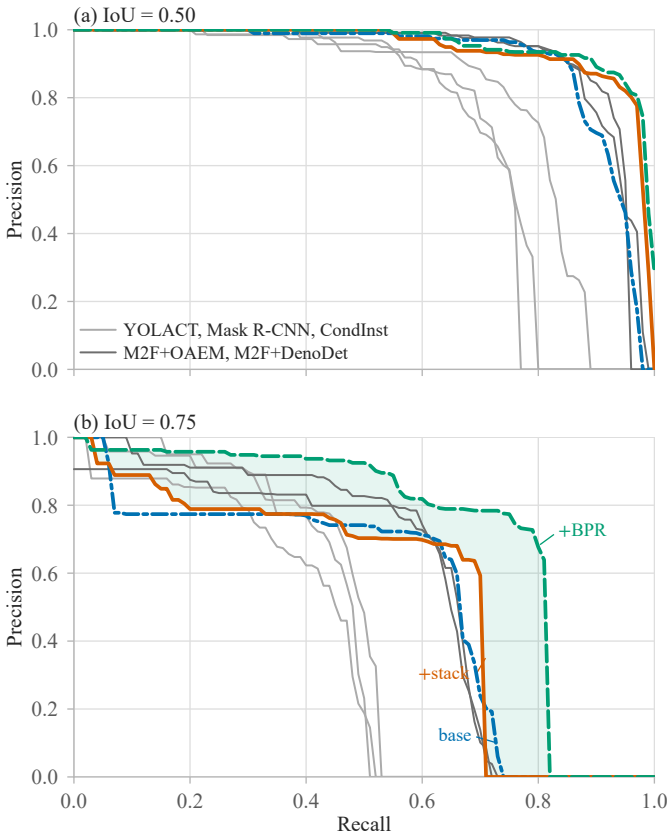


Fig. 4. Precision-recall curves on the PSeg-SSDD inshore split, recomputed from the Table II dumps before the learned re-ranking, at (a) IoU = 0.50 and (b) IoU = 0.75. Our three variants are labelled at the curve ends; the five competitors form generic (light gray) and SAR-specific (dark gray) tiers, named in (a)'s legend. The shaded band in (b) is the refiner's gap. Per-method AP: Table II. Offshore curves are omitted (saturated).

E. Ablation Study

1) Progressive Recovery Ladder: Table IV takes the cascade apart one operator per row: soup, six-view TTA,

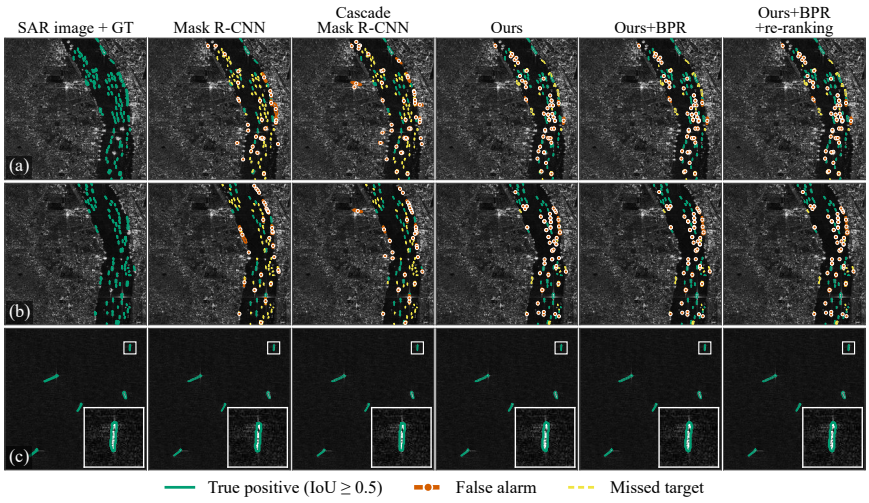


Fig. 5. Qualitative comparison on three HRSID test scenes (0.3 score display threshold, mask IoU ≥ 0.5 matching). Green solid contours are true positives, orange dash-dot contours with a centroid dot false alarms, and yellow dashed contours missed targets. Columns: SAR image with ground truth, Mask R-CNN, Cascade Mask R-CNN, the training-free stack, + boundary-patch refiner, + learned re-ranker. Rows (a) and (b) are inshore harbors; row (c) is offshore, with an inset on the most disputed hull.

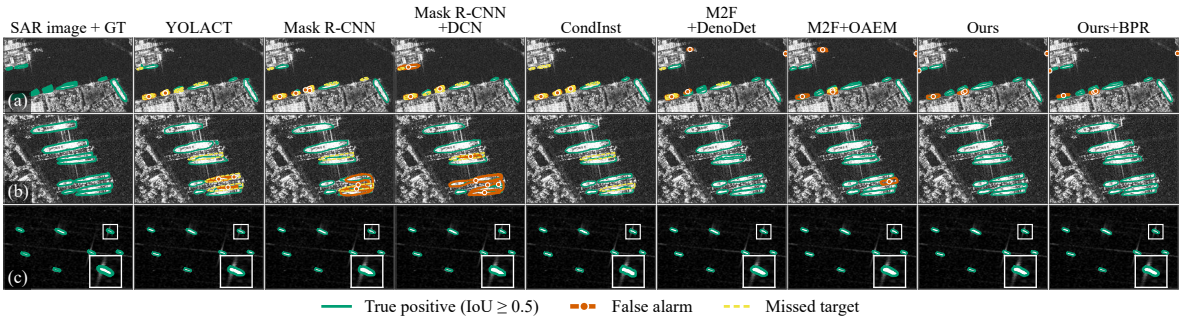


Fig. 6. Qualitative comparison of the eight PSeg-SSDD methods of Table II on three test scenes (0.3 score display threshold, mask IoU ≥ 0.5 matching; color coding as in Fig. 5; column names printed in the figure). Rows (a) and (b) are inshore scenes; row (c) is offshore, where every method detects every ship and the inset magnifies the hull with the largest inter-method mask-IoU spread.

TABLE IV
PROGRESSIVE RECOVERY LADDER ON HRSID (MASK AP $\times 100$). EACH ROW ADDS ONE OPERATOR TO THE PREVIOUS ROW; NO ROW RETRAINS OR MODIFIES THE BASE. FULL COCO METRICS ARE REPORTED FOR THE THREE AUDITED ENDPOINTS

			Inshore			Official (full)			Offshore		
Configuration	Train. cost	Infer. cost	AP	AP_{50}	AP_{75}	AP	AP_{50}	AP_{75}	AP	AP_{50}	AP_{75}
c0 base, COCO-init M2F-R50 (seed s_0)	55 ep	$1\times$	27.7	61.8	22.1	45.6	81.9	49.4	59.6	97.2	71.2
+ weight soup (avg. of s0/s2/s3)	0	$1\times$	29.6	–	–	47.4	–	–	61.1	–	–
+ 6-view multi-scale TTA $\{736, 800, 864\}\times$flip	0	$\sim 6\times$	36.1	–	–	52.5	–	–	64.4	–	–
+ wide 8-view TTA $\{640, 800, 960, 1088\}\times$flip, box-NMS	0	$\sim 8\times$	37.1	–	–	52.6	–	–	64.2	–	–
+ consensus-vote fusion (full method: carriers + operator)	0	$\sim 8\times$	38.8	75.6	36.1	53.8	89.4	62.2	64.8	98.5	81.7
+ BPR refiner (trained corrector, Sec. IV-E3)	19 min	$\sim 8\times + \varepsilon$	43.6	76.3	47.0	58.6	89.9	70.7	69.1	**98.6**	87.4
+ learned re-ranking (trained corrector, Sec. IV-E3)	19 s	$\sim 8\times + \varepsilon$	**44.5**	**76.7**	**48.3**	**59.4**	**90.1**	**71.4**	**70.1**	**98.6**	**88.0**

Bold denotes the best value in each column; ties at one decimal are both marked. "–": full metrics are logged only for the three audited endpoints. The upper block is the training-free full method; the two lower rows are the optional trained correctors.

widened eight-view TTA, and consensus-vote fusion compose a lift of +11.1/+8.2/+5.2 inshore/official/offshore over the 27.7/45.6/59.6 s_0 base at zero additional training and roughly $8\times$ inference. The six-view TTA gain holds at +6.9/+7.4/+6.7 across the three seeds (mean +7.0, s.d. 0.3), so the recovery is not a single-seed artifact. The first two operators compose near-additively: soup alone contributes +1.9 inshore and six-view TTA alone +6.9, predicting 36.5 where the measured combination is 36.1, and we claim no synergy. The member count is not a monotone knob either. Table V sweeps N from 1 to 6 on the PSeg-SSDD base, adding seeds in a fixed order. The $N = 2$ and $N = 5$ soups fall below every one of their members on all three splits. The $N = 3$ and $N = 4$ soups exceed every member offshore (64.8 and 65.5 against a member maximum of 64.3) and the six-seed official mean of 60.7 (60.9 and 61.6). Inshore they sit below the six-

TABLE V
WEIGHT SOUP VERSUS MEMBER COUNT N ON PSEG-SSDD (MASK AP $\times 100$, SINGLE-VIEW BASE, NO TTA). MEMBERS ARE ADDED CUMULATIVELY IN THE FIXED SEED ORDER $s_0, s_2, s_3, s_4, s_5, s_6$

N	Members	Inshore	Official	Offshore
1	s_0	53.1	60.2	63.3
2	s_0, s_2	48.3	57.2	61.6
3	s_0, s_2, s_3	51.9	60.9	64.8
4	s_0, s_2, s_3, s_4	52.3	61.6	65.5
5	s_0, s_2, s_3, s_4, s_5	45.2	55.9	61.4
6	$s_0, s_2, s_3, s_4, s_5, s_6$	48.0	59.4	64.2
Six single seeds, mean $\pm$ s.d.		52.9 $\pm$ 0.7	60.7 $\pm$ 0.3	64.1 $\pm$ 0.4

TABLE VI
FUSION OPERATORS ON IDENTICAL SOUP $\times$ 8-VIEW CANDIDATES (MASK AP $\times 100$)

Fusion operator	Inshore	Official	Offshore
box-NMS (reference)	37.1	52.6	64.2
matrix-NMS	4.1	8.9	14.4
majority pixel vote	**39.1**	48.7	55.9
probability mean	**39.1**	48.7	55.9
seed vote	38.2	53.5	**64.8**
consensus vote ($S_{\max} = 6$)	38.8	**53.8**	**64.8**

Bold denotes the best value in each column.
Band-ceiling sweep on the same candidates: $S_{\max} = 4/5/6$ gives inshore 38.4/38.6/38.8, offshore pinned at 64.8 throughout.

seed mean of 52.9$\pm$0.7 (51.9 and 52.3). The three members used throughout were fixed by the seed budget before this sweep, not selected from it. Pairwise soups of the same six checkpoints explain the pattern: the seeds fall into two groups, $\{s_0, s_3, s_4, s_6\}$ and $\{s_2, s_5\}$. All seven within-group pairs beat the mean of their two members on every split (inshore +0.3 to +0.8, official +0.8 to +1.4, offshore +0.8 to +1.8). All eight cross-group pairs lose on the inshore and official splits, by 5.0 to 12.7 inshore points. Parameter space does not separate the groups: every pair sits at an L_2 distance of 10.0 to 10.8 (against $\|\theta_{s_0}\| = 300.3$), and the cosine between fine-tuning increments is 0.46 to 0.52. The three-member PSeg-SSDD soup straddles the groups, which is why it sits below the single-seed mean, and the rise and fall of Table V tracks the share of cross-group members. The two conditions of Section III-B are therefore necessary, not sufficient. On the 55-epoch HRSID schedule the same three seeds are compatible (soup +1.9 inshore, matching the ensemble control below), and a training-split compatibility test is left to future work. The fusion operator contributes +1.7/+1.2/+0.6 over box-NMS on the identical candidate set; Section IV-E2 shows this term is an enabler, not a third additive gain. As a control, a three-model prediction ensemble over the same eight views reaches 39.3/53.6/64.0 at $3\times$ model cost against 38.8/53.8/64.8 for the soup-based method at $1\times$, so the soup captures the seed diversity of a prediction ensemble at one third of its model cost [9].

2) Fusion Operator: Every operator in Table VI consumes the identical soup $\times$ eight-view candidate set; only the merge rule changes, and the fuser decides whether aggressive TTA helps or harms. Box-NMS suppresses in box-IoU space, where berthed parallel hulls share high box IoU while their masks barely overlap, so hard deletion removes true inshore neighbors and caps the reference at 37.1. Matrix-NMS [37] deletes nothing, so each object arrives roughly eight times, the decayed near-duplicates flood the top-100 ranking, and AP collapses to 4.1/8.9/14.4. Majority pixel voting is the strongest inshore reading at 39.1 but erodes offshore to 55.9, below the 59.6 no-TTA base, because multi-scale views misalign by one to two pixels along long hull boundaries and unconditional voting trims every disputed pixel; the probability-mean control reproduces it bitwise, so the correct degree of freedom is which mask to keep, not how to average pixels. The seed vote, which keeps each cluster's highest-scoring mask verbatim, never erodes but concedes inshore at 38.2. The consensus vote resolves the trade-off by gating on cross-view agreement: it matches or exceeds the seed vote on every split (38.8/53.8/64.8; inshore AP_{75} 36.1 against 35.3) and is monotone in the band ceiling, $S_{\max} = 4/5/6$ yielding inshore 38.4/38.6/38.8 with offshore pinned at 64.8 throughout. The view count matters less than the fuser. Under the same consensus fusion, two views ($\{800\}\times$flip) give 33.6/50.5/63.4, four views ($\{800, 960\}\times$flip) 38.3/53.8/64.9, and eight views 38.8/53.8/64.8; four views already land within 0.5 inshore point of eight.

3) The Two Trained Correctors: The lower block of Table IV isolates the two correctors. The refiner adds +4.8/+4.7/+4.4 AP inshore/official/offshore; these are differences before rounding, and the rounded table entries differ by at most 0.1. The paired bootstrap of Section IV-C puts this gain at $[+4.05, +5.55]$ inshore (+4.76, 369 images), $[+4.25, +5.22]$ official (+4.72, 1962 images), and $[+3.81, +4.89]$ offshore (+4.39, 1593 images), positive in all 1000 resamples. Its metric fingerprint is exactly that of a boundary correction: inshore AP_{50} moves by only +0.7 while AP_{75} moves by +10.9 (+8.4 official, +5.7 offshore). AP_s rises from 35.8 to 41.4 inshore, 52.8 to 57.8 official, and 64.1 to 68.7 offshore. The gain therefore concentrates at strict overlap on the small near-shore hulls that the stride-4 halo diagnosis of Section III-C targeted. Because a corrector trained on (prediction, ground-truth) pairs invites a leakage suspicion, the application path was audited along seven attack vectors, all clean: the apply path receives no annotation and its output regenerates MD5-identically from predictions, refiner weights, and images alone; zero edited pixels fall outside the $\pm$4-pixel band in 90 of 90 sampled instances, 0 of the roughly 237 000 detection scores (three splits combined) change, and the filename overlap between refiner training patches and test images is zero. The band is a formal guarantee, not a tuning knob. The same refiner applied with $r \in \{2, 4, 8\}$ pixels, or with no band at all, returns 43.6/58.6/69.1 in all four cases (differences below 0.01 AP): it writes only near the boundary anyway. Two further refiner training seeds give 43.1/58.1/68.7 and 43.2/58.3/68.8, a three-seed mean of 43.3$\pm$0.2/58.3$\pm$0.2/68.9$\pm$0.2. The seed-0 refiner reported throughout is the best of the three by 0.3 to 0.5 point, and the re-ranking figures below build on it.

Re-ranking is worth a further +0.9/+0.8/+1.0 AP as tabulated (+0.97/+0.82/+0.95 before rounding): small and positive at every threshold, as a pure re-ordering of a fixed detection

set must be. AP_s reaches 42.0/58.3/69.4. One column moves the other way and we report it plainly: large-object AP falls by 1.4, 1.3, and 1.3 points, because the regressor is fitted on training candidates that are overwhelmingly small vessels, and on a large-vessel-dominated corpus the blend would need re-examination. Its audit mirrors the refiner's: the regressor is fitted only on training-split predictions and annotations, every feature is computable from the prediction alone, and not one mask byte changes. A paired percentile bootstrap over test images (Section IV-C) gives a 95% confidence interval of $[+0.61, +1.23]$ for the inshore AP gain (point estimate +0.97, 369 images). On the official split the interval is $[+0.61, +1.00]$ (+0.82, 1962 images) and offshore $[+0.63, +1.18]$ (+0.95, 1593 images). The gain is positive in every one of the 1000 resamples on all three splits.

4) Architecture- and Training-Side Controls: Table VII records the regime experiment: seven architecture- and training-side directions attempted on the same strong COCO-initialized base under matched protocol, none of which survives as a contribution. OffsetAdj constrains the deformable-attention sampling offsets that COCO pretraining has already tuned and costs 1.0 inshore. CHART, a one-to-many auxiliary supervision, collapsed at auxiliary weight 1.0 and was flat at 0.1. CAST hard-sample pasting and the four training-side families (capacity, feature enrichment, data augmentation, supervision) were absorbed by the strong baseline. The mask-IoU rescoring head is worth +1.5 inshore standalone at its $\lambda = 0.2$ operating point yet leaves a +0.1 residual under the stack, subsumed by the soup. The SGA strip adapter [28], which preserves the COCO initialization behind a zero-initialized residual gate [25], lost 0.9 inshore while gaining 0.6 offshore. Swin-S [38] carries a +3.2 COCO mask-AP advantage over ResNet-50 that shrinks to +0.9 inshore on SAR, the only positive base-level delta in the sweep. Under the recovery stack (single seed, no soup) it ends below the ResNet-50 method on all three splits. The 200-epoch equal-budget control saturated near epoch 45. Its best official base of 46.4, reached with an early-decayed warm restart, is +0.8 concentrated in AP_{75}, with the checkpoint selected on the official test split, which favors the control. Under the stack it ends 0.9/1.3/1.3 below the ResNet-50 method, so longer training neither closes the inshore gap nor compensates for the missing soup.

TABLE VII
ARCHITECTURE- AND TRAINING-SIDE CONTROLS (MASK AP $\times$100). UPPER BLOCK: VERSUS THE C0 BASE; LOWER BLOCK: VERSUS THE FULL METHOD

Configuration	Inshore	Official	Offshore
c0 base (3-seed reference)*	27.7	45.7	59.8
OffsetAdj (constrained offsets, 1 seed)	26.7	45.3	59.7
CHART (aux. supervision, w=0.1, 1 seed)	≈27.7	45.7	59.8
CAST (hard-sample paste augm., 1 seed)	27.7	–	–
Training-side families (4)	flat (subsumed by base)		
Mask-IoU head (λ=0.2, seed s_0)	29.2	47.3	62.0
SGA (zero-gated strip adapter, 1 seed)	26.8	45.5	60.4
Swin-S backbone swap (1 seed)	28.6	46.1	59.9
Swin-S + recovery stack (1 seed)	37.3	52.5	63.5
200 ep + recovery stack (1 seed)	37.9	52.5	63.5
R-50 full method (reference)	**38.8**	**53.8**	**64.8**

Bold denotes the best value in each column. The seven directions are OffsetAdj, CHART, CAST, the four training-side families, the mask-IoU head, SGA, and the Swin-S swap; the 200-epoch row is a training-budget control. *Official/offshore means of the 3-seed control axis, within the $\pm$0.3 noise band of the seed-s_0 figures 45.6/59.6 of Table IV.

F. Cross-Dataset Transfer and Universality

1) Transfer to PSeg-SSDD: The recipe transfers to PSeg-SSDD with one proportional change, the TTA scale set of Section IV-B; the consensus band is carried over unchanged, and the two correctors are refitted on SSDD training-split predictions with no operator retuned. The lower block of Table II gives the ladder: from the 51.9/60.9/64.8 soup base to 62.1/67.5/69.8 on the inshore/official/offshore splits, a composed lift of +10.2/+6.6/+5.0. The tiers trade places as the regime changes. SSDD single-view predictions are more stable than their HRSID counterparts (base inshore 51.9 against 27.7), so the variance-driven stack has less to remove, +2.1 inshore against +11.1. The boundary tier instead dominates, contributing +6.0 of the +10.2 and raising inshore AP_{75} from 54.9 to 72.5, which re-ranking carries to 74.6; official-split AP_{75} moves from 82.9 to 84.9. The paired bootstrap is wider here. On the 46 inshore images, base to stack gives +2.02, $[-0.78, +4.36]$ (positive in 93% of resamples), stack to refiner +6.09, $[+2.06, +9.30]$, and refiner to re-ranker +2.09, $[-0.09, +3.87]$ (97%). On the 186 offshore images all three steps are positive: $[+0.93, +3.23]$, $[+0.74, +2.81]$, and $[+0.21, +1.68]$. Two comparisons with O2Former [2] bound the result. Its from-scratch Mask2Former baseline on this dataset is reported at 52.5 inshore, above our 51.9 soup base at 36 epochs. That soup itself sits below its members, whose six single seeds average 52.9$\pm$0.7 inshore (Table V). The regime gap of Section IV-D therefore does not reappear on PSeg-SSDD, and the lead here comes from the inference-side cascade. Under the same split protocol, the complete cascade closes to 1.1 inshore and 2.1 offshore points below the 500-epoch specialized architecture itself (63.2/71.9).

2) Universality Across Frozen Carriers: Everything so far rides on one carrier. Table VIII therefore hands the identical recipe to six further carrier–dataset pairs across five distinct architectures, with no per-carrier tuning; each carrier is a single frozen training run without soup. All 66 cells come from re-evaluation, and every one of the 21 endpoints improves over its own carrier: on HRSID, +11.9 inshore apiece for Mask R-CNN and Cascade R-CNN and +14.3 for SOLOv2; on PSeg-SSDD, +6.9, +16.3, and +6.1. Two intermediate readings are negative and we print them: the training-free rung costs Mask R-CNN 0.6 and YOLACT 0.5 offshore AP on PSeg-SSDD, the erosion Section IV-E2 identified for predictions that are already stable, and the boundary tier converts both to +1.4 and +2.7. The most useful number in the table is 42.3: Cascade R-CNN, a 2018 architecture with no SAR-specific component, lands within 0.8 point of the 43.1 that a specialized 500-epoch design reports on the same inshore split, purely by looking eight times and correcting itself. These carriers are our own trainings under our own budget, so what transfers to Table I is the delta, not the level.

The variation of the gain tests the diagnosis. Across the

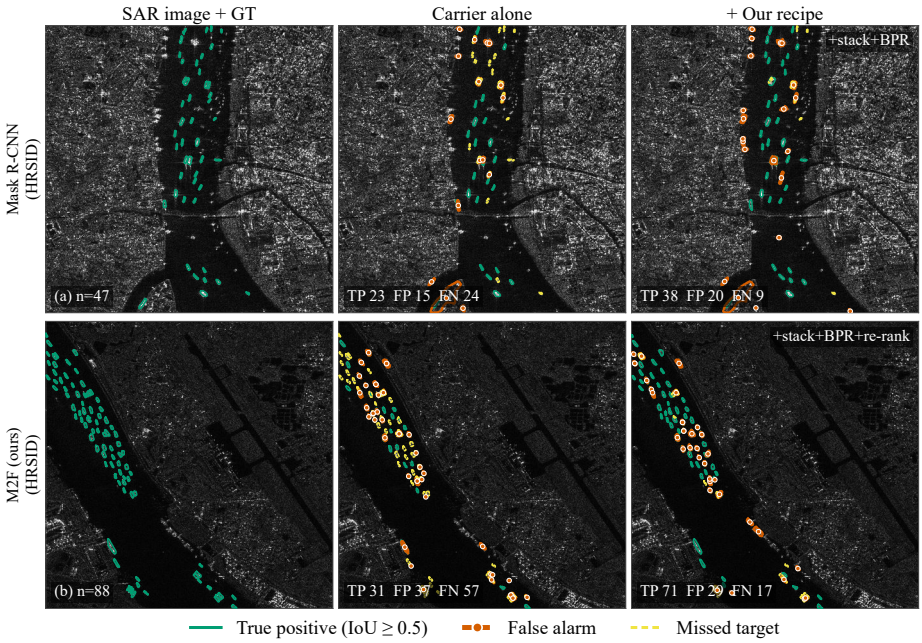


Fig. 7. The identical inference-side recipe applied to two frozen carriers on the HRSID inshore split (Table VIII). Columns: SAR image with ground truth, carrier alone, carrier with the recipe applied (0.3 score display threshold; color coding as in Fig. 5; detection counts printed in each panel). Rows: (a) Mask R-CNN with the boundary-refined stack; (b) our Mask2Former carrier, which also applies the learned re-ranker.

TABLE VIII
UNIVERSALITY OF THE INFERENCE-SIDE RECIPE ACROSS FROZEN CARRIERS (MASK AP ×100)

Carrier	Variant	Inshore	Official	Offshore
HRSID				
Mask R-CNN	carrier alone	28.3	44.8	57.8
	+stack	37.1	52.2	63.6
	+BPR	40.2	56.2	68.2
Cascade R-CNN	carrier alone	30.4	46.5	59.2
	+stack	39.1	53.3	64.0
	+BPR	42.3	57.6	69.0
SOLOv2	carrier alone	19.0	38.2	54.0
	+stack	25.5	43.0	56.3
	+BPR	33.3	53.0	67.6
M2F (ours)	carrier alone	29.7	47.6	61.4
	+stack	38.8	53.8	64.8
	+BPR	43.6	58.6	69.1
	+re-rank	44.5	59.4	70.1
PSeg-SSDD				
Mask R-CNN	carrier alone	40.8	57.8	65.1
	+stack	44.1	58.6	64.5
	+BPR	47.7	61.3	66.5
CondInst	carrier alone	41.2	54.3	60.1
	+stack	50.1	59.7	63.6
	+BPR	57.5	66.1	69.3
YOLACT	carrier alone	38.9	55.7	63.4
	+stack	41.1	56.2	62.9
	+BPR	45.0	59.7	66.1

Each carrier is trained once and then frozen. "+stack" applies the wide eight-view multi-scale flip TTA with consensus-vote fusion, "+BPR" further applies a per-carrier boundary-patch refiner, and "+re-rank" further applies the learned re-ranker. Official is the full official test set of each dataset; inshore and offshore are its two scene subsets. The M2F carrier-alone row is the weight soup with the λ=0.2 mask-IoU rescoring that the stack configuration carries (Section IV-E4), hence 29.7/47.6/61.4 instead of the 29.6/47.4/61.1 of Table IV.

seven carriers, the Spearman rank correlation between carrier-alone inshore AP and inshore gain is -0.77 without CondInst ($n = 6$) and -0.11 with it ($n = 7$). Neither is significant at these sample sizes. The exception has a mechanism: CondInst's per-instance kernels are the most view-sensitive readout of the seven. Its excess gain sits in the variance tier (+8.9 inshore from the stack against +3.3 and +2.2 for the other PSeg-SSDD carriers). On the three HRSID carriers the two error axes scale in opposite directions: the stack returns +8.8 and +8.7 inshore on the two region-based carriers against +6.5 on SOLOv2, whereas the refiner returns +3.1 and +3.2 against +7.8, because SOLOv2's grid-predicted kernels yield the loosest silhouettes of the three (carrier-alone inshore AP_{75} 15.7 against 24.8 and 27.0). The two dependences carry opposite signs, which no single-axis account of the near-shore residual can produce. Figure 7 shows the deltas on the image, with the counts printed in each panel: in the 47-ship HRSID harbor of row (a), the recipe takes Mask R-CNN from 23 to 38 found targets. In the 88-ship harbor of row (b) it takes our own carrier from 31 to 71 true positives while the false alarms fall from 37 to 29. Recall and precision improve together, which a lowered threshold cannot do.

G. What Did Not Help

Four further directions came back null or negative, and each duplicates a remedy already in place. First, speckle marginalization: a despeckle-and-redraw variant reached 37.1 inshore and a direct mild-Gamma variant 37.3, against 38.8 for the geometric stack. The decisions a speckle re-draw can flip lie inside the set the 640-to-1088-pixel views already flip. In both variants, fusion that keeps the base prediction as arbiter beat pure draw-majority voting (37.1 against 27.0,

37.3 against 30.0). Second, a miss-rescue verifier that re-scores the rejected candidate tail (scores in $[0.01, 0.35)$) moved the three splits to 43.5/58.5/69.1 against 43.6/58.6/69.1. The dumps already run down to score zero at 100 detections per image, so nothing is missing; a badly placed detection can only be promoted, which the re-ranker does. Third, extending the re-ranker's 14 features with two decoder-depth-consistency descriptors ties at 44.5/59.3/70.0 against 44.5/59.4/70.1, a post-hoc comparison, not a selection step. Stability across decoder depth and agreement across views carry the same information here. Fourth, a minimal-footprint variant of the base improved every split of both datasets (28.2/46.3/60.5 on HRSID against 27.7/45.6/59.6) and then surrendered all of it under the stack, 36.4/51.6/63.0 against 38.8/53.8/64.8. A training-side gain and an inference-side gain on the same variance axis substitute for one another.

V. Discussion

A. A Preservation Regularity: When Architecture Engineering Pays

One regularity organizes the controlled ledger of Section IV: on a strong COCO-pretrained base, a modification survives in proportion to how much of the pretrained representation it preserves. Every weight-touching intervention of Table VII failed to pay under the recovery stack; the carriers plus the fusion operator delivered the entire +11.1/+8.2/+5.2 lift. The regularity is regime-dependent. On a from-scratch Mask2Former at 17.9 inshore mask AP, the specialized modules of O2Former are worth +25.2 points [2]. In our own earlier experiments, a Gaussian–Wasserstein matching cost gained +4.4 inshore on a from-scratch-head base (12.5 to 16.9; single-seed base, two-seed mean). The same cost turned neutral to negative once the identical model was COCO-initialized (27.9 to 27.4, single seed). Architecture engineering pays where the representation leaves headroom for injected priors. Preservation is necessary but not sufficient. The strip adapter respected it through a zero-initialized gate and still lost 0.9 inshore point, because the elongation prior mismatches the quasi-point-like near-shore hulls of HRSID. The two correctors obey the same regularity from outside the network.

B. Limitations

Five limitations bound our claims. First, single-look amplitude imagery imposes a sensor ceiling: the physics that separates a berthed hull from the dock structure it touches is absent from the input. A radiometric-consistency re-ranking probe accordingly recovered +0.0 inshore over a flip-TTA control. Second, the residual gap has moved, not closed. Offshore HRSID still trails O2Former by 1.6 points (70.1 against 71.7), and PSeg-SSDD remains 1.1 inshore and 2.1 offshore below the same architecture. Third, the evidence base is two collections of single-sensor amplitude imagery. The sweep of Table VII is eliminative rather than exhaustive, so stronger interventions than the forms we tested may exist. Fourth, the training-free claim is scoped to the recovery stack of Section III. Each corrector is refitted on the target training split, and the refined-tier comparison inherits the operating-point caveat of Section IV. Fifth, the soup depends on member compatibility: the six PSeg-SSDD seeds average well within two groups and poorly across them (Section IV-E1). A training-split test that picks compatible members without touching test data is left to future work.

VI. Conclusion

This article proposed a three-tier self-correcting inference cascade for SAR ship instance segmentation, together with a controlled study of the pretraining regime it rests on. To recover the detections single-view inference loses in near-shore clutter, we introduced a consensus-gated mask-IoU vote fusion operator and deployed it on a standard weight-soup and wide-TTA carrier. Its ablation showed that the operator, not the view count, decides whether aggressive augmentation helps or collapses. To move the two residuals the stack leaves, we devised two small correctors supervised only by the frozen model's own training-split errors. A band-guarded boundary-patch refiner edits only mask pixels, and a learned consensus re-ranker edits only scores; both were audited for leakage. Experiments on HRSID and PSeg-SSDD demonstrated the effectiveness and robustness of the cascade: 44.5 inshore mask AP and 48.3 inshore AP_{75} on HRSID, and 62.1 inshore AP on PSeg-SSDD. On the HRSID near-shore split these figures exceed the strongest specialized 500-epoch architecture at one third of its epochs and roughly $8\times$ test-time cost. Ablation experiments underlined that architecture- and training-side modifications of the strong base were flat, negative, or absorbed. The identical recipe (stack plus refiner), in contrast, lifted six further carrier–dataset pairs across five distinct architectures on every split. In the future, we plan to extend the recipe to sensors providing polarimetric or sub-aperture information, where the amplitude ceiling identified here may be broken.

Data and Code Availability

The HRSID and PSeg-SSDD datasets are publicly available from their original authors. Our code, all training and inference configurations, and the evaluation scripts will be released at https://github.com/XXX/XXX upon acceptance. The release includes the COCO-format annotations, the inshore/offshore split lists with the scripts that produce them, and the souped Mask2Former-R50, refiner, and re-ranker checkpoints for both datasets. The implementation builds on MMDetection 3.3.0 and PyTorch 2.0.1.

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